

Code Localization as Sequential Set-Coverage: A Trainable Multi-Round Scaffolding for Small LLMs

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Abstract

Before a coding agent can fix a GitHub issue, it must localize the files and functions to change; a mistake here cannot be undone by any later reasoning. Strong localization scaffoldings such as OpenHands and SWE-Agent are built for long contexts and large models, so the open 7–14B code models most teams can run overflow a 32K-token budget before the search ends. Training small models is no easier: a multi-step agent yields only a sparse terminal reward, forcing expensive reinforcement learning. We present a multi-round localization scaffolding that removes both barriers. The agent refines a fixed-length candidate list over rounds, each round inspecting a bounded window of ranked nodes and their neighborhood on a code graph, so per-call token cost stays roughly constant. The round structure also exposes a dense per-round target: under potential-based shaping, the per-round signal telescopes to end-to-end recall@ L . The agent therefore trains by ordinary supervised fine-tuning—a weighted cross-entropy with a differentiable coverage term—rather than sparse-reward RL. This recasts localization from ranking to sequential set-coverage on a code graph. Trained this way, small open models become competitive with GPT Codex 5.3 under the same scaffolding, and a 30B model matches open-ended OpenHands agents at a fraction of the token cost, on SWE-Bench Lite, SWE-Bench Verified, and the multilingual SWE-PolyBench.

1 Introduction

LLM coding agents increasingly resolve real GitHub issues by editing a repository directly, and benchmarks such as SWE-Bench now track how often they succeed (Jimenez et al., 2024b). A patch can only be correct, however, if it changes the right code. Before any edit, the agent must identify the files and functions an issue actually touches, the step known as *code localization*. This step is decisive and largely unrecoverable: once a wrong

location enters the agent’s editable context, the repair stage reasons over the wrong code, and no later reasoning recovers the correct patch (Jimenez et al., 2024b; Yang et al., 2024a). The strongest systems are built around this fact. Xia et al. (2024a) drop agentic interaction and resolve issues with a fixed localize-then-repair pipeline, crediting much of their accuracy to a hierarchical localization phase, and methods that improve the localization stage report corresponding gains in end-to-end resolution (Sohrabizadeh et al., 2025). Localization is therefore not a preprocessing convenience but the step that determines how well downstream repair can perform.

Yet localizing well is hard: a real repository holds thousands of files joined by tangled import, call, and inheritance edges, while an issue report describes symptoms in natural language that can sit far from the code that must change. Current systems close this gap in four ways. Reinforcement learning methods train the search policy itself: RepoSearcher pairs tool-integrated RL with rejection-sampled supervised fine-tuning (Ma et al., 2025), and SWE-RL scales RL over open software-evolution data (Wei et al., 2025). General-purpose agent scaffoldings give a capable LLM an open-ended action space with shell access, as in SWE-Agent and OpenHands (Yang et al., 2024a; Wang et al., 2025). Graph-exploration agents such as LocAgent let an LLM traverse a repository code graph, hopping freely between related code units (Chen et al., 2025b). Dense retrieval methods skip the agent altogether: SWERank scores code units with a bi-encoder retriever and a listwise LLM reranker (Reddy et al., 2025b).

The first three families pay for accuracy in tokens, and none of them bounds the cost. An RL agent, an open-ended scaffold, and a graph explorer all keep issuing tool calls and appending observations until they choose to stop, so a localization trajectory grows without a predictable ceiling. Under

an OpenHands-style scaffolding, a single localization run can exceed 131K tokens (Section 4.4). A model with a long context window and ample compute absorbs that cost; the open 7–14B code models most teams can realistically run do not. A Qwen-class coder with a 32K-token window overflows before the search terminates. Reinforcement learning compounds the problem: a multi-step episode returns only a sparse terminal reward, so adapting a small model to it takes many long and expensive rollouts. Dense retrieval is the one cheap option, but a single-step, order-centric ranker cannot revisit a discarded candidate or follow a structural edge, so it leaves the sequential, graph-structured side of the problem untouched. No existing method is at once bounded in token cost at inference and cheap enough to train on a small open model.

2 Related Work

Reinforcement learning for localization. RepoSearcher equips a repository agent with a curated tool set and trains it through rejection-sampled supervised fine-tuning followed by tool-integrated RL, reaching strong localization with a 32B model (Ma et al., 2025). SWE-RL scales RL over open software-evolution data and shows the learned reasoning transfers across software tasks (Wei et al., 2025). RepoNavigator narrows the agent to a single jump-to-definition tool and trains end-to-end with RL from a base model, so that a 7B agent can outperform substantially larger baselines (Zhang et al., 2025). In every case training optimizes multi-step episodes that return only a sparse terminal reward, so adaptation is costly in rollouts and compute. Our round decomposition exposes a dense per-round target that telescopes to recall@L (Section 3.3), turning the same problem into ordinary supervised fine-tuning.

Agentic scaffolding. SWE-Agent and OpenHands wrap a strong LLM in a general-purpose environment with shell access and an open-ended action space, and they set much of the standard for end-to-end issue resolution (Yang et al., 2024a; Wang et al., 2025); Moatless provides a comparable tool-driven loop (Örwall, 2024). Agentless instead drops interaction and runs a fixed localize-then-repair pipeline, showing that a simple prompted procedure can rival full agents (Xia et al., 2024a). The interactive scaffoldings place no ceiling on trajectory length, so token cost is unbounded and a localization run easily overflows the context window

of the open 7–14B models we target. Agentless is bounded, but its localization is a fixed prompted pipeline, not a component a small model can learn to improve. Our scaffolding bounds the token cost of every round and stays trainable, so a small model can run the search within budget and be trained to localize better.

Graph-based exploration. A related line hands the LLM the repository’s structure. LocAgent builds a heterogeneous code graph and lets the model explore it through graph-traversal actions (Chen et al., 2025b), CoSIL has the model search a call graph it expands on the fly without a pre-built index (Jiang et al., 2025), and RepoGraph supplies a repository-level code graph for software-engineering agents (Ouyang et al., 2025). This work establishes that structural edges carry localization signal. The traversal, however, is unconstrained, so trajectory length and token cost again grow without a bound. We use the same structure through bounded neighborhood expansion around ranked seeds, so graph signal raises recall without inflating the budget.

Retrieval and reranking. Localization is also posed as retrieval. Lexical matching (Robertson et al., 1994) and learned code embeddings (Feng et al., 2020; Guo et al., 2021; Zhang et al., 2024a) score code units against the issue text, and SWERank sharpens this for issue localization with a bi-encoder retriever and a listwise LLM reranker (Reddy et al., 2025b); CoRet improves the retriever for code-editing queries (Fehr et al., 2025) and SpIDER incorporates repository structure into the dense representation (Chaudhari et al., 2025). These rankers are cheap and bounded, but the objective is per-query and order-centric, optimizing the order of one list rather than coverage of the gold set, and a single pass cannot revisit a discarded unit or follow a structural edge. SWERank+ adds a multi-turn agentic loop with a memory buffer (Reddy et al., 2025a), yet the loop is open-ended and the training signal stays a ranking objective. We recast localization as sequential set-coverage on a code graph and train an objective provably co-optimal with recall@L (Section 3.4), keeping the bounded cost of retrieval while adding the set recovery and multi-round refinement that ranking omits.

3 Method

3.1 Problem Setup

3.2 Multi-Round Localization Scaffolding

3.3 A Dense, Recall-Aligned Per-Round Target

3.4 Supervised Training Objective

4 Experiments

4.1 Setup

4.2 Main Results

4.3 Does the Coverage Term Help?

4.4 Efficiency and Scaling

5 Conclusion

Limitations

References

Toufique Ahmed, Premkumar Devanbu, Christoph Treude, and Michael Pradel. 2024. Can llms replace manual annotation of software engineering artifacts? *arXiv preprint arXiv:2408.05534*.

Nadia Alshahwan, Jubin Chheda, Anastasia Finogenova, Beliz Gokkaya, Mark Harman, Inna Harper, Alexandru Marginean, Shubho Sengupta, and Eddy Wang. 2024. Automated unit test improvement using large language models at meta. In *Companion Proceedings of the 32nd ACM International Conference on the Foundations of Software Engineering*, pages 185–196.

Ben Athiwaratkun, Sanjay Krishna Gouda, Zijian Wang, Xiaopeng Li, Yuchen Tian, Ming Tan, Wasi Uddin Ahmad, Shiqi Wang, Qing Sun, Mingyue Shang, Sujan Kumar Gonugondla, Hantian Ding, Varun Kumar, Nathan Fulton, Arash Farahani, Siddhartha Jain, Robert Giaquinto, Haifeng Qian, Murali Krishna Ramanathan, and 6 others. 2023. [Multi-lingual evaluation of code generation models](#). In *The Eleventh International Conference on Learning Representations*.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and 1 others. 2021. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

Loubna Ben Allal, Niklas Muennighoff, Logesh Kumar Umapathi, Ben Lipkin, and Leandro von Werra. 2022. A framework for the evaluation of code generation models. <https://github.com/bigcode-project/bigcode-evaluation-harness>.

Shravan Chaudhari, Rahul Thomas Jacob, Mononito Goswami, Jiajun Cao, Shihab Rashid, and Christian Bock. 2025. SpIDER: Spatially informed dense

embedding retrieval for software issue localization. *arXiv preprint arXiv:2512.16956*.

Jiayi Chen and 1 others. 2025a. Training software engineering agents and verifiers with SWE-gym. *arXiv preprint arXiv:2412.21139*.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde De Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, and 1 others. 2021. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*.

Zhaoling Chen, Robert Tang, Gangda Deng, Fang Wu, Jialong Wu, Zhiwei Jiang, Viktor Prasanna, Arman Cohan, and Xingyao Wang. 2025b. [LocAgent: Graph-guided LLM agents for code localization](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8697–8727, Vienna, Austria. Association for Computational Linguistics.

Neil Chowdhury, James Aung, Chan Jun Shern, Oliver Jaffe, Dane Sherburn, Giulio Starace, Evan Mays, Rachel Dias, Marwan Aljube, Mia Glaese, Carlos E. Jimenez, John Yang, Leyton Ho, Tejal Patwardhan, Kevin Liu, and Aleksander Madry. 2024a. [Introducing SWE-bench Verified](#). Accessed on March 2, 2025.

Neil Chowdhury, James Aung, Chan Jun Shern, Oliver Jaffe, Dane Sherburn, Giulio Starace, Evan Mays, Rachel Dias, Marwan Aljube, Mia Glaese, Carlos E. Jimenez, John Yang, Kevin Liu, and Aleksander Madry. 2024b. [Introducing SWE-bench Verified](#). OpenAI Blog.

Colin Clement, Dawn Drain, Jonathan Timcheck, Alexey Svyatkovskiy, and Neel Sundaresan. 2020. [PyMT5: multi-mode translation of natural language and python code with transformers](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9052–9065, Online. Association for Computational Linguistics.

Jianbo Dai, Jianqiao Lu, Yunlong Feng, Rongju Ruan, Ming Cheng, Haochen Tan, and Zhijiang Guo. 2024. Mhpp: Exploring the capabilities and limitations of language models beyond basic code generation. *arXiv preprint arXiv:2405.11430*.

Chunyu Deng, Yilun Zhao, Xiangru Tang, Mark Gerson, and Arman Cohan. 2023. Benchmark probing: Investigating data leakage in large language models. In *NeurIPS 2023 Workshop on Backdoors in Deep Learning-The Good, the Bad, and the Ugly*.

Yangruibo Ding, Zijian Wang, Wasi Ahmad, Murali Krishna Ramanathan, Ramesh Nallapati, Parminder Bhatia, Dan Roth, and Bing Xiang. 2024. [CoCoMIC: Code completion by jointly modeling in-file and cross-file context](#). In *Proceedings of the 2024 Joint*

288	<i>International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)</i> , pages 3433–3445, Torino, Italia.	
289	ELRA and ICCL.	
290		
291		
292	Yangruibo Ding, Zijian Wang, Wasi Uddin Ahmad, Hantian Ding, Ming Tan, Nihal Jain, Murali Krishna Ramanathan, Ramesh Nallapati, Parminder Bhatia, Dan Roth, and Bing Xiang. 2023. Crosscodeeval: A diverse and multilingual benchmark for cross-file code completion . In <i>Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track</i> .	
293		
294		
295		
296		
297		
298		
299		
300	Fabio James Fehr, Prabhu Teja S, Luca Franceschi, and Giovanni Zappella. 2025. CoRet: Improved retriever for code editing . In <i>Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)</i> , pages 775–789, Vienna, Austria. Association for Computational Linguistics.	
301		
302		
303		
304		
305		
306		
307	Zhangyin Feng, Daya Guo, Duyu Tang, Nan Duan, Xiaocheng Feng, Ming Gong, Linjun Shou, Bing Qin, Ting Liu, Daxin Jiang, and Ming Zhou. 2020. CodeBERT: A pre-trained model for programming and natural languages . In <i>Findings of the Association for Computational Linguistics: EMNLP 2020</i> , pages 1536–1547, Online. Association for Computational Linguistics.	
308		
309		
310		
311		
312		
313		
314		
315	Paul Gauthier. 2024. Aider is ai pair programming in your terminal. https://github.com/paul-gauthier/aider .	
316		
317		
318	Charles Goodhart. 1976. <i>Monetary relationships: a view from Threadneedle Street</i> . University of Warwick.	
319		
320		
321	Alex Gu, Baptiste Rozière, Hugh Leather, Armando Solar-Lezama, Gabriel Synnaeve, and Sida I Wang. 2024. Cruxeval: A benchmark for code reasoning, understanding and execution. <i>arXiv preprint arXiv:2401.03065</i> .	
322		
323		
324		
325		
326	Sumit Gulwani, Oleksandr Polozov, Rishabh Singh, and 1 others. 2017. Program synthesis. <i>Foundations and Trends® in Programming Languages</i> , 4(1-2):1–119.	
327		
328		
329	Daya Guo, Shuo Ren, Shuai Lu, Zhangyin Feng, Duyu Tang, Shujie Liu, Long Zhou, Nan Duan, Alexey Svyatkovskiy, Shengyu Fu, Michele Tufano, Shao Kun Deng, Colin Clement, Dawn Drain, Neel Sundaresan, Jian Yin, Daxin Jiang, and Ming Zhou. 2021. Graphcodebert: Pre-training code representations with data flow . <i>Preprint</i> , arXiv:2009.08366.	
330		
331		
332		
333		
334		
335		
336	Daya Guo, Canwen Xu, Nan Duan, Jian Yin, and Julian McAuley. 2023. Longcoder: A long-range pre-trained language model for code completion. In <i>International Conference on Machine Learning</i> , pages 12098–12107. PMLR.	
337		
338		
339		
340		
341	Jiafeng Guo, Yixing Fan, Qingyao Ai, and W. Bruce Croft. 2016. A deep relevance matching model for ad-hoc retrieval . CIKM ’16, page 55–64, New York, NY, USA. Association for Computing Machinery.	
342		
343		
344		
	Jiafeng Guo, Yixing Fan, Liang Pang, Liu Yang, Qingyao Ai, Hamed Zamani, Chen Wu, W. Bruce Croft, and Xueqi Cheng. 2020. A deep look into neural ranking models for information retrieval . In <i>Information Processing & Management</i> , 57(6):102067.	345
		346
		347
		348
		349
	Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Song, and Jacob Steinhardt. 2021. Measuring coding challenge competence with APPS . In <i>Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)</i> .	350
		351
		352
		353
		354
		355
		356
	Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models . In <i>International Conference on Learning Representations</i> .	357
		358
		359
		360
		361
	Qian Huang, Jian Vora, Percy Liang, and Jure Leskovec. 2024. Mlagentbench: evaluating language agents on machine learning experimentation. In <i>Proceedings of the 41st International Conference on Machine Learning</i> , ICML’24. JMLR.org.	362
		363
		364
		365
		366
	Hamel Husain, Ho-Hsiang Wu, Tiferet Gazit, Miltiadis Allamanis, and Marc Brockschmidt. 2019. CodeSearchNet challenge: Evaluating the state of semantic code search. <i>arXiv preprint arXiv:1909.09436</i> .	367
		368
		369
		370
	Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. 2024. Livecodebench: Holistic and contamination free evaluation of large language models for code. <i>arXiv preprint arXiv:2403.07974</i> .	371
		372
		373
		374
		375
		376
	Naman Jain, Skanda Vaidyanath, Arun Iyer, Nagarajan Natarajan, Suresh Parthasarathy, Sriram Rajamani, and Rahul Sharma. 2022. Jigsaw: large language models meet program synthesis . In <i>Proceedings of the 44th International Conference on Software Engineering</i> , ICSE ’22, page 1219–1231, New York, NY, USA. Association for Computing Machinery.	377
		378
		379
		380
		381
		382
		383
	Juyong Jiang, Fan Wang, Jiasi Shen, Sungju Kim, and Sunghun Kim. 2024. A survey on large language models for code generation. <i>arXiv preprint arXiv:2406.00515</i> .	384
		385
		386
		387
	Zhonghao Jiang, Xiaoxue Ren, Meng Yan, Wei Jiang, Yong Li, and Zhongxin Liu. 2025. Cosil: Software issue localization via llm-driven code repository graph searching . <i>Preprint</i> , arXiv:2503.22424.	388
		389
		390
		391
	Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R Narasimhan. 2024a. Can language models resolve real-world github issues? https://www.swebench.com/ .	392
		393
		394
		395
		396
	Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R Narasimhan. 2024b. SWE-bench: Can language models resolve real-world github issues? In <i>The</i>	397
		398
		399
		400

401	<i>Twelfth International Conference on Learning Representations.</i>	456
402		457
403	Liqliang Jing, Zhehui Huang, Xiaoyang Wang, Wen-	458
404	lin Yao, Wenhao Yu, Kaixin Ma, Hongming Zhang,	459
405	Xinya Du, and Dong Yu. 2024. Dsbench: How far	
406	are data science agents to becoming data science ex-	
407	perts? <i>Preprint</i> , arXiv:2409.07703.	
408	James A Jones, Mary Jean Harrold, and John Stasko.	
409	2002. Visualization of test information to assist fault	
410	localization. In <i>International Conference on Soft-</i>	
411	<i>ware Engineering</i> , pages 467–477.	
412	Sungmin Kang, Gabin An, and Shin Yoo. 2024. A quan-	
413	titative and qualitative evaluation of llm-based ex-	
414	plainable fault localization. <i>Proc. ACM Softw. Eng.</i> ,	
415	1(FSE).	
416	Yuhang Lai, Chengxi Li, Yiming Wang, Tianyi Zhang,	
417	Ruiqi Zhong, Luke Zettlemoyer, Wen-Tau Yih,	
418	Daniel Fried, Sida Wang, and Tao Yu. 2023. DS-	
419	1000: A natural and reliable benchmark for data sci-	
420	ence code generation. In <i>Proceedings of the 40th</i>	
421	<i>International Conference on Machine Learning</i> , vol-	
422	ume 202 of <i>Proceedings of Machine Learning Re-</i>	
423	<i>search</i> , pages 18319–18345. PMLR.	
424	Jia Li, Ge Li, Xuanming Zhang, Yihong Dong, and	
425	Zhi Jin. 2024a. Evocodebench: An evolving code	
426	generation benchmark aligned with real-world code	
427	repositories. <i>arXiv preprint arXiv:2404.00599</i> .	
428	Jia Li, Ge Li, Yunfei Zhao, Yongmin Li, Huanyu Liu,	
429	Hao Zhu, Lecheng Wang, Kaibo Liu, Zheng Fang,	
430	Lanshen Wang, Jiazheng Ding, Xuanming Zhang,	
431	Yuqi Zhu, Yihong Dong, Zhi Jin, Binhua Li, Fei	
432	Huang, Yongbin Li, Bin Gu, and Mengfei Yang.	
433	2024b. DevEval: A manually-annotated code gener-	
434	ation benchmark aligned with real-world code repos-	
435	itories. In <i>Findings of the Association for Comput-</i>	
436	<i>ational Linguistics ACL 2024</i> , pages 3603–3614,	
437	Bangkok, Thailand and virtual meeting. Association	
438	for Computational Linguistics.	
439	Ryan Li, Yanzhe Zhang, and Diyi Yang. 2024c. Sketch2code: Evaluating vision-language models	
440	for interactive web design prototyping. <i>Preprint</i> ,	
441	arXiv:2410.16232.	
442		
443	Yi Li, Shaohua Wang, and Tien N. Nguyen. 2021. Fault	
444	localization with code coverage representation learn-	
445	ing. In <i>International Conference on Software Engi-</i>	
446	<i>neering</i> .	
447	Yujia Li, David Choi, Junyoung Chung, Nate Kush-	
448	man, Julian Schrittwieser, Rémi Leblond, Tom Ec-	
449	cles, James Keeling, Felix Gimeno, Agustin Dal	
450	Lago, Thomas Hubert, Peter Choy, Cyprien de Mas-	
451	son d’Autume, Igor Babuschkin, Xinyun Chen, Po-	
452	Sen Huang, Johannes Welbl, Sven Gowal, Alexey	
453	Cherepanov, and 7 others. 2022. Competition-	
454	level code generation with alphacode. <i>Science</i> ,	
455	378(6624):1092–1097.	
	Tianyang Liu, Canwen Xu, and Julian McAuley. 2024. Repobench: Benchmarking repository-level code	
	auto-completion systems. In <i>The Twelfth Interna-</i>	
	<i>tional Conference on Learning Representations</i> .	
	LocAgent Team. 2025. LocAgent: Graph-guided	
	LLM agents for code localization. <i>arXiv preprint</i>	
	<i>arXiv:2503.09089</i> . TODO: verify author list.	
	Qinyu Luo, Yining Ye, Shihao Liang, Zhong Zhang,	
	Yujia Qin, Yaxi Lu, Yesai Wu, Xin Cong, Yankai	
	Lin, Yingli Zhang, Xiaoyin Che, Zhiyuan Liu, and	
	Maosong Sun. 2024. Repoagent: An llm-powered	
	open-source framework for repository-level code doc-	
	umentation generation. <i>Preprint</i> , arXiv:2402.16667.	
	Zexiong Ma, Chao Peng, Qunhong Zeng, Pengfei Gao,	
	Yanzhen Zou, and Bing Xie. 2025. Tool-integrated	
	reinforcement learning for repo deep search. <i>arXiv</i>	
	<i>preprint arXiv:2508.03012</i> .	
	Zohar Manna and Richard J. Waldinger. 1971. To-	
	ward automatic program synthesis. <i>Commun. ACM</i> ,	
	14(3):151–165.	
	Samuel Miserendino, Michele Wang, Tejal Patwardhan,	
	and Johannes Heidecke. 2025. Swe-lancer: Can fron-	
	tier llms earn 1 million from real-world freelance soft-	
	ware engineering? <i>arXiv preprint arXiv:2502.12115</i> .	
	Andrew Y Ng, Daishi Harada, and Stuart Russell. 1999.	
	Policy invariance under reward transformations: The-	
	ory and application to reward shaping. In <i>Interna-</i>	
	<i>tional Conference on Machine Learning</i> , pages 278–	
	287.	
	Erik Nijkamp, Bo Pang, Hiroaki Hayashi, Lifu Tu, Huan	
	Wang, Yingbo Zhou, Silvio Savarese, and Caiming	
	Xiong. 2023. Codegen: An open large language	
	model for code with multi-turn program synthesis. In	
	<i>The Eleventh International Conference on Learning</i>	
	<i>Representations</i> .	
	OrcaLoca Team. 2025. OrcaLoca: An LLM agent	
	framework for software issue localization. <i>arXiv</i>	
	<i>preprint</i> . TODO: verify authors and arXiv id.	
	Siru Ouyang, Wenhao Yu, Kaixin Ma, Zilin Xiao, Zhi-	
	han Zhang, Mengzhao Jia, Jiawei Han, Hongming	
	Zhang, and Dong Yu. 2025. Repograph: Enhancing	
	AI software engineering with repository-level code	
	graph. In <i>The Thirteenth International Conference</i>	
	<i>on Learning Representations</i> .	
	Mike Papadakis and Yves Le Traon. 2015. Metallaxis-	
	FL: Mutation-based fault localization. <i>Software Test-</i>	
	<i>ing, Verification and Reliability</i> , 25(5-7):605–628.	
	Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan	
	Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang,	
	Bill Qian, and 1 others. 2024. ToolLLM: Facilitating	
	large language models to master 16000+ real-world	
	APIs. <i>arXiv preprint arXiv:2307.16789</i> .	

508	Muhammad Shihab Rashid, Christian Bock, Yuan	Marc Szafraniec, Baptiste Roziere, Hugh James Leather,	562
509	Zhuang, Alexander Buchholz, Tim Esler, Si-	Patrick Labatut, Francois Charton, and Gabriel Syn-	563
510	mon Valentin, Luca Franceschi, Martin Wistuba,	naeve. 2023. Code translation with compiler repre-	564
511	Prabhu Teja Sivaprasad, Woo Jung Kim, Anoop	sentations . In <i>The Eleventh International Conference</i>	565
512	Deoras, Giovanni Zappella, and Laurent Callot.	<i>on Learning Representations</i> .	566
513	2025. Swe-polybench: A multi-language bench-		
514	mark for repository level evaluation of coding agents .	Runchu Tian, Yining Ye, Yujia Qin, Xin Cong, Yankai	567
515	<i>Preprint</i> , arXiv:2504.08703.	Lin, Yinxu Pan, Yesai Wu, Hui Haotian, Liu We-	568
		ichuan, Zhiyuan Liu, and Maosong Sun. 2024. De-	569
516	Revanth Gangi Reddy, Ye Liu, Wenting Zhao, Jae-	bugBench: Evaluating debugging capability of large	570
517	Hyeok Doo, Tarun Suresh, Daniel Lee, Caiming	language models . In <i>Findings of the Association for</i>	571
518	Xiong, Yingbo Zhou, Semih Yavuz, and Shafiq	<i>Computational Linguistics ACL 2024</i> , pages 4173–	572
519	Joty. 2025a. SweRank+: Multilingual, multi-turn	4198, Bangkok, Thailand and virtual meeting. Asso-	573
520	code ranking for software issue localization. <i>arXiv</i>	ciation for Computational Linguistics.	574
521	<i>preprint arXiv:2512.20482</i> .		
522	Revanth Gangi Reddy, Tarun Suresh, JaeHyeok Doo,	Mark Vero, Niels Mündler, Victor Chibotaru, Veselin	575
523	Ye Liu, Xuan Phi Nguyen, Yingbo Zhou, Semih	Raychev, Maximilian Baader, Nikola Jovanović,	576
524	Yavuz, Caiming Xiong, Heng Ji, and Shafiq Joty.	Jingxuan He, and Martin Vechev. 2025. Baxbench:	577
525	2025b. Swerank: Software issue localization with	Can llms generate correct and secure backends?	578
526	code ranking . <i>Preprint</i> , arXiv:2505.07849.		
527	Stephen E. Robertson, Steve Walker, Susan Jones,	Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu,	579
528	Micheline Hancock-Beaulieu, and Mike Gatford.	Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi	580
529	1994. Okapi at trec-3 . In <i>Text Retrieval Conference</i> .	Song, Bowen Li, Jaskirat Singh, Hoang H. Tran,	581
		Fuqiang Li, Ren Ma, Mingzhang Zheng, Bill Qian,	582
530	Moshe Shaked and J. George Shanthikumar. 2007.	Yanjun Shao, Niklas Muennighoff, Yizhe Zhang,	583
531	<i>Stochastic Orders</i> . Springer.	Binyuan Hui, and 5 others. 2025. Openhands: An	584
		open platform for AI software developers as gener-	585
532	Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu,	alist agents . In <i>The Thirteenth International Confer-</i>	586
533	Junxiao Song, Mingchuan Zhang, Y.K. Li, Y. Wu,	<i>ence on Learning Representations</i> .	587
534	and Daya Guo. 2024. DeepSeekMath: Pushing the	Xingyao Wang and 1 others. 2024. OpenHands: An	588
535	limits of mathematical reasoning in open language	open platform for AI software developers as general-	589
536	models. <i>arXiv preprint arXiv:2402.03300</i> .	ist agents. <i>arXiv preprint arXiv:2407.16741</i> .	590
537	Haiyang Shen, Yue Li, Desong Meng, Dongqi Cai,	Zhiruo Wang, Grace Cuenca, Shuyan Zhou, Frank F Xu,	591
538	Sheng Qi, Li Zhang, Mengwei Xu, and Yun Ma. 2025.	and Graham Neubig. 2022. Mconala: a benchmark	592
539	Shortcutsbench: A large-scale real-world benchmark	for code generation from multiple natural languages.	593
540	for api-based agents . <i>Preprint</i> , arXiv:2407.00132.	<i>arXiv preprint arXiv:2203.08388</i> .	594
541	Atefeh Sohrabizadeh, Jialin Song, Mingjie Liu, Rajarshi	Yuxiang Wei, Olivier Duchenne, Jade Copet, Quentin	595
542	Roy, Chankyu Lee, Jonathan Raiman, and Bryan	Carbonneaux, Lingming Zhang, Daniel Fried,	596
543	Catanzaro. 2025. Nemotron-CORTEXA: Enhancing	Gabriel Synnaeve, Rishabh Singh, and Sida I. Wang.	597
544	LLM agents for software engineering tasks via im-	2025. SWE-RL: Advancing LLM reasoning via	598
545	proved localization and solution diversity . In <i>Forty-</i>	reinforcement learning on open software evolution.	599
546	<i>second International Conference on Machine Learn-</i>	<i>arXiv preprint arXiv:2502.18449</i> .	600
547	<i>ing</i> .	Yonghao Wu, Zheng Li, Jie M. Zhang, Mike Pa-	601
548	Stack Overflow. 2023. Stack over-	padakis, Mark Harman, and Yong Liu. 2023. Large	602
549	flow developer survey 2023. https:	language models in fault localisation . <i>Preprint</i> ,	603
550	//survey.stackoverflow.co/2023/	arXiv:2308.15276.	604
551	#most-popular-technologies-language-prof .	Chunqiu Steven Xia, Yinlin Deng, Soren Dunn, and	605
552	Tarun Suresh, Revanth Gangi Reddy, Yifei Xu, Zach	Lingming Zhang. 2024a. Agentless: Demystify-	606
553	Nussbaum, Andriy Mulyar, Brandon Duderstadt, and	ing LLM-based software engineering agents. <i>arXiv</i>	607
554	Heng Ji. 2025. CoRNStack: High-quality contrastive	<i>preprint arXiv:2407.01489</i> .	608
555	data for better code retrieval and reranking . In <i>The</i>	Chunqiu Steven Xia, Yinlin Deng, Soren Dunn, and	609
556	<i>Thirteenth International Conference on Learning</i>	Lingming Zhang. 2024b. Agentless: Demystify-	610
557	<i>Representations</i> .	ing llm-based software engineering agents . <i>Preprint</i> ,	611
		arXiv:2407.01489.	612
558	SweRank Team. 2025. SweRank: Software issue	Yiqing Xie, Alex Xie, Divyanshu Sheth, Pengfei Liu,	613
559	localization with code ranking. <i>arXiv preprint</i>	Daniel Fried, and Carolyn Rose. 2024. Codebench-	614
560	<i>arXiv:2505.07849</i> . TODO: verify author list and	gen: Creating scalable execution-based code genera-	615
561	title.	tion benchmarks. <i>arXiv preprint arXiv:2404.00566</i> .	616

617	John Yang, Carlos E Jimenez, Alexander Wettig, Kilian	world repo-level coding challenges. <i>arXiv preprint</i>	674
618	Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir	<i>arXiv:2401.07339</i> .	675
619	Press. 2024a. SWE-agent: Agent-computer inter-		
620	faces enable automated software engineering. <i>arXiv</i>	Yuge Zhang, Qiyang Jiang, XingyuHan XingyuHan,	676
621	<i>preprint arXiv:2405.15793</i> .	Nan Chen, Yuqing Yang, and Kan Ren. 2024d.	677
622		Benchmarking data science agents . In <i>Proceedings</i>	678
623	John Yang, Carlos E Jimenez, Alexander Wettig, Kil-	<i>of the 62nd Annual Meeting of the Association for</i>	679
624	ian Lieret, Shunyu Yao, Karthik Narasimhan, and	<i>Computational Linguistics (Volume 1: Long Papers)</i> ,	680
625	Ofir Press. 2024b. Swe-agent: Agent-computer inter-	pages 5677–5700, Bangkok, Thailand. Association	681
626	faces enable automated software engineering. <i>arXiv</i>	for Computational Linguistics.	682
627	<i>preprint arXiv:2405.15793</i> .		
628	John Yang, Carlos E. Jimenez, Alex L. Zhang, Kil-	Zhaoxi Zhang, Yitong Duan, Yanzhi Zhang, Yiming Xu,	683
629	ian Lieret, Joyce Yang, Xindi Wu, Ori Press,	Zhixiang Wang, Kun Liang, Yang Li, Jiahui Liang,	684
630	Niklas Muennighoff, Gabriel Synnaeve, Karthik R.	Deguo Xia, Jizhou Huang, Jiyan He, and Yunfang	685
631	Narasimhan, Diyi Yang, Sida I. Wang, and Ofir	Wu. 2025. One tool is enough: Reinforcement learn-	686
632	Press. 2024c. Swe-bench multimodal: Do ai systems	ing for repository-level LLM agents. <i>arXiv preprint</i>	687
633	generalize to visual software domains? <i>Preprint</i> ,	<i>arXiv:2512.20957</i> .	688
634	arXiv:2410.03859.		
635	Hao Yu, Bo Shen, Dezhi Ran, Jiaxin Zhang, Qi Zhang,	Mingchen Zhuge, Changsheng Zhao, Dylan Ashley,	689
636	Yuchi Ma, Guangtai Liang, Ying Li, Qianxiang Wang,	Wenyi Wang, Dmitrii Khizbullin, Yunyang Xiong,	690
637	and Tao Xie. 2024. Codereval: A benchmark of prag-	Zechun Liu, Ernie Chang, Raghuraman Krishnamoor-	691
638	matic code generation with generative pre-trained	thi, Yuandong Tian, and 1 others. 2024. Agent-as-a-	692
639	models . In <i>Proceedings of the IEEE/ACM 46th Inter-</i>	judge: Evaluate agents with agents. <i>arXiv preprint</i>	693
640	<i>national Conference on Software Engineering, ICSE</i>	<i>arXiv:2410.10934</i> .	694
641	'24, New York, NY, USA. Association for Computing		
642	Machinery.	Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu,	695
643	Daoguang Zan, Bei Chen, Dejian Yang, Zeqi Lin, Minsu	Wenhao Yu, Ratnadira Widayarsi, Imam Nur Bani	696
644	Kim, Bei Guan, Yongji Wang, Weizhu Chen, and	Yusuf, Haolan Zhan, Junda He, Indraneil Paul, and	697
645	Jian-Guang Lou. 2022. Cert: continual pre-training	1 others. 2024. Bigcodebench: Benchmarking code	698
646	on sketches for library-oriented code generation.	generation with diverse function calls and complex	699
647	<i>arXiv preprint arXiv:2206.06888</i> .	instructions. <i>arXiv preprint arXiv:2406.15877</i> .	700
648			
649	Daoguang Zan, Zhirong Huang, Wei Liu, Hanwu Chen,	Albert Örwall. 2024. Moatless tools .	701
650	Linhao Zhang, Shulin Xin, Lu Chen, Qi Liu, Xiao-		
651	jian Zhong, Aoyan Li, Siyao Liu, Yongsheng Xiao,	A Proof of the Telescoping Identity	702
652	Liangqiang Chen, Yuyu Zhang, Jing Su, Tianyu Liu,	B Scaffolding and Training Details	703
653	Rui Long, Kai Shen, and Liang Xiang. 2025. Multi-	C Additional Results	704
654	swe-bench: A multilingual benchmark for issue re-	D AI Disclosure	705
655	solving . <i>Preprint</i> , arXiv:2504.02605.		
656			
657	Daoguang Zan, Zhirong Huang, Ailun Yu, Shaoxin Lin,		
658	Yifan Shi, Wei Liu, Dong Chen, Zongshuai Qi, Hao		
659	Yu, Lei Yu, Dezhi Ran, Muhan Zeng, Bo Shen, Pan		
660	Bian, Guangtai Liang, Bei Guan, Pengjie Huang, Tao		
661	Xie, Yongji Wang, and Qianxiang Wang. 2024. Swe-		
662	bench-java: A github issue resolving benchmark for		
663	java . <i>Preprint</i> , arXiv:2408.14354.		
664			
665	Dejiao Zhang, Wasi Uddin Ahmad, Ming Tan, Hantian		
666	Ding, Ramesh Nallapati, Dan Roth, Xiaofei Ma, and		
667	Bing Xiang. 2024a. CODE REPRESENTATION		
668	LEARNING AT SCALE . In <i>The Twelfth Interna-</i>		
669	<i>tional Conference on Learning Representations</i> .		
670			
671	Dejiao Zhang, Wasi Uddin Ahmad, Ming Tan, Hantian		
672	Ding, Ramesh Nallapati, Dan Roth, Xiaofei Ma, and		
673	Bing Xiang. 2024b. CODE REPRESENTATION		
	LEARNING AT SCALE . In <i>The Twelfth Interna-</i>		
	<i>tional Conference on Learning Representations</i> .		
	Kechi Zhang, Jia Li, Ge Li, Xianjie Shi, and Zhi		
	Jin. 2024c. Codeagent: Enhancing code gener-		
	ation with tool-integrated agent systems for real-		